FHRP Configuration Guide

First Published: 2025-09-15

Last Modified: 2026-07-17

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CONTENTS

CHAPTER 1

HSRP

Feature history for HSRP

This table provides release and platform support information for the features explained in this module.

These features are available in all the releases subsequent to the one they were introduced in, unless noted otherwise.

Release	Feature name and description	Supported platform
Cisco IOS XE 26.2.1ea	HSRP: HSRP feature support has been introduced.	Cisco C9550 Series Smart Switches
Cisco IOS XE 17.18.1	HSRP: HSRP is the standard method used by Cisco to provide high network availability through first-hop redundancy for IP hosts on an IEEE 802 LAN configured with a default gateway IP address.	Cisco C9350 Series Smart Switches Cisco C9610 Series Smart Switches

Understand HSRP

HSRP is the standard method used by Cisco to provide high network availability through first-hop redundancy for IP hosts on an IEEE 802 LAN configured with a default gateway IP address. HSRP routes IP traffic without relying on the availability of any single router. It enables a set of router interfaces to work together to present the appearance of a single virtual router or default gateway to the hosts on a LAN. When HSRP is configured on a network or segment, it provides a virtual Media Access Control (MAC) address and an IP address that is shared among a group of configured routers.

Two or more routers configured with HSRP can use the MAC address and IP network address of a virtual router. The virtual router does not exist; it represents the common target for routers that are configured to provide backup to each other. The active router is selected among the routers, with another designated as the standby router. This standby router assumes control of the group's MAC and IP address if the active router fails.

Note

- Routers in an HSRP group can include routed ports and switch virtual interfaces (SVIs), as long as they support HSRP.
- On Cisco 9610 Smart Series Switches, HSRP is not supported on subinterfaces.

HSRP benefits

HSRP provides high-network availability by ensuring redundancy for IP traffic from hosts on networks. In a group of router interfaces, the active router is the router of choice for routing packets; the standby router is the router that takes over the routing duties when an active router fails or when preset conditions are met.

HSRP is useful for hosts that do not support a router discovery protocol and cannot switch to a new router when their selected router reloads or loses power. When HSRP is configured on a network segment, it provides a virtual MAC address and an IP address that is shared among router interfaces in a group of router interfaces running HSRP. The router selected by the protocol to be the active router receives and routes packets destined for the group's MAC address. For a given number 'n' of routers running HSRP, there are 'n +1' IP and MAC addresses assigned.

HSRP detects when the designated active router fails, and a selected standby router assumes control of the MAC and IP addresses of the Hot Standby group. A new standby router is also selected at that time. Devices running HSRP send and receive multicast, UDP-based 'hello' packets to detect a router failure and to designate active and standby routers.

When HSRP is configured on interfaces, ICMP redirect messages are automatically enabled. ICMP is a network layer Internet protocol that provides message packets for reporting errors and information relevant to IP processing. ICMP provides diagnostic functions. It sends and directs error packets to the host. This feature filters outgoing ICMP redirect messages through HSRP, changing the next hop IP address to an HSRP virtual IP address.

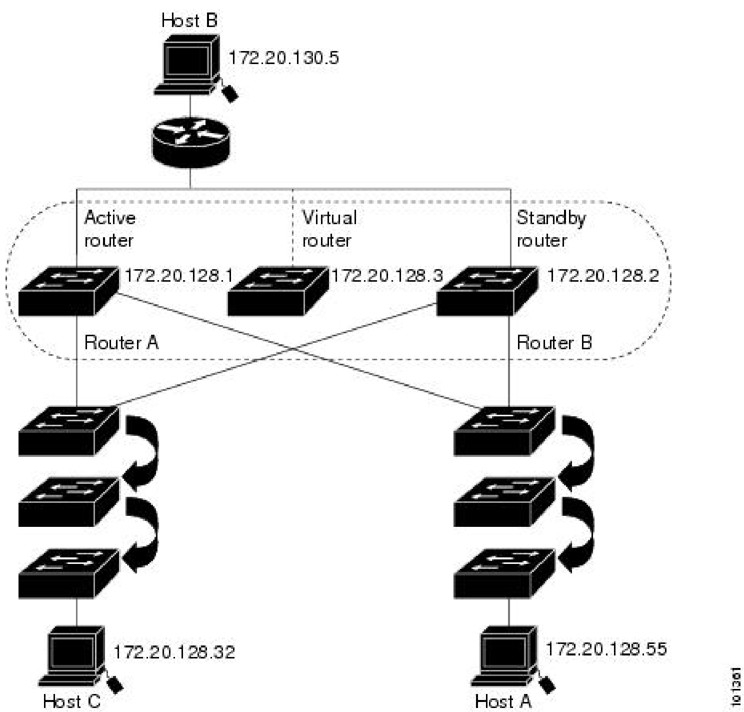
HSRP operation

You can configure multiple Hot Standby groups among switches and switch stacks operating in Layer 3 to coordinate the use of redundant routers. To do so, specify a group number for each Hot Standby command group you configure for an interface. For example, you might configure an interface on switch 1 as an active router and one on switch 2 as a standby router and also configure another interface on switch 2 as an active router with another interface on switch 1 as its standby router.

The figure illustrates a network segment configured for HSRP. Each router is configured with the MAC address and IP address of the virtual router. Instead of configuring hosts on the network with the IP address of Router A, you configure them with the IP address of the virtual router as their default router. When Host C sends packets to Host B, it directs them to the MAC address of the virtual router. If Router A stops transferring packets for any reason, Router B assumes the active router duties by responding to the virtual IP address and MAC address. Host C continues to use the IP address of the virtual router to address packets destined for Host

B, which Router B now receives and sends to Host B. Until Router A resumes operation, HSRP allows Router B to provide uninterrupted service to users on Host C's segment that need to communicate with users on Host B's segment. It also continues to handle packets between the Host A segment and Host B, fulfilling its normal function.

Figure 1: Typical HSRP configuration



HSRP versions

The switch supports these HSRP versions:

- HSRPv1: Version 1 is the default version and has these features:
  - The HSRP group number can range from 0 to 255.
  - HSRPv1 uses the multicast address 224.0.0.2 to send hello packets, which can conflict with Cisco Group Management Protocol (CGMP) leave processing. You cannot enable HSRPv1 and CGMP at the same time; they are mutually exclusive.
- HSRPv2: Version 2 has these features:
  - HSRPv2 uses the multicast address 224.0.0.102 to send hello packets. HSRPv2 and CGMP leave processing are not mutually exclusive, so you can enable both simultaneously.
  - HSRPv2 has a different packet format than HRSPv1.

When running HSRPv1, the switch cannot identify the physical router that sends a hello packet because the router's source MAC address appears as the virtual MAC address.

HSRPv2 has a different packet format than HSRPv1. An HSRPv2 packet uses the type-length-value (TLV) format and includes a 6-byte identifier field with the MAC address of the physical router that sent the packet.

If an interface running HSRPv1 gets an HSRPv2 packet, the type field is ignored.

HSRP for IPv6 operation

HSRP ensures routing redundancy for IPv6 traffic without depending on a single router's availability. IPv6 hosts learn about available routers through IPv6 neighbor discovery router advertisement messages that are multicast periodically or solicited by hosts.

An HSRP IPv6 group features a virtual MAC address derived from the HSRP group number and a virtual IPv6 link-local address, which is, by default, derived from the HSRP virtual MAC address. The range is 0005.73A0.0000 through 0005.73A0.0FFF (4096 addresses).

When the HSRP group is active, it sends periodic messages related to the HSRP virtual IPv6 link-local address. The messaging ceases after a final message is sent when the group exits the active state.

Note To configure HSRP for IPv6, ensure that HSRP version 2 (HSRPv2) is enabled on the interface.

HSRP group IPv6 operation

A link-local address is an IPv6 unicast address that can be automatically configured on any interface using the link-local prefix FE80::/10 (1111 1110 10) and the interface identifier in the modified EUI-64 format. Link-local addresses are used in the stateless autoconfiguration process. Nodes on a local link can communicate using link-local addresses. They do not require site-local or globally unique addresses.

When you enter the **standby ipv6** command, a link-local address is generated from the link-local prefix, and a modified EUI-64 format interface identifier is generated in which the EUI-64 interface identifier is created from the relevant HSRP virtual MAC address.

In IPv6, a device on the link uses RA messages to advertise any site-local and global prefixes and its capability to function as a default device for the link. RA messages are sent periodically and in response to router solicitation messages, sent by hosts during system startup.

A node on the link can automatically configure site-local and global IPv6 addresses by appending its interface identifier (64 bits) to the prefixes (64 bits) included in the RA messages. The resulting 128-bit IPv6 addresses that are configured by the node are then subjected to duplicate address detection to ensure their uniqueness on the link. If the prefixes advertised in the RA messages are globally unique, then the IPv6 addresses configured by the node are also guaranteed to be globally unique. Router solicitation messages, which have a value of 133 in the Type field of the ICMP packet header, are sent by hosts at system startup for immediate autoconfiguration and avoiding the wait for the next scheduled RA message.

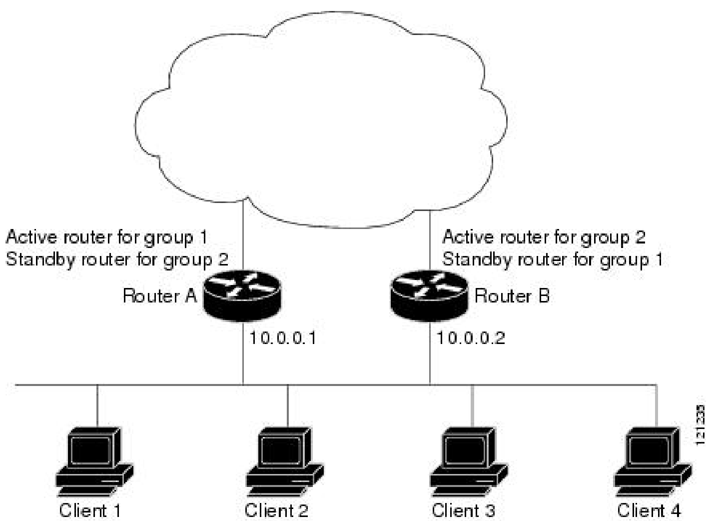
Multiple HSRP

The switch supports Multiple HSRP (MHSRP), an extension of HSRP that enables load sharing across multiple HSRP groups. Configure MHSRP to achieve load-balancing and to use two or more standby groups (and paths) from a host network to a server network.

In the figure, half of the clients are configured for Router A, and the other half for Router B. The configurations of Routers A and B together establish two HSRP groups. For group 1, Router A is the active router due to its higher priority, with Router B as standby. For group 2, Router B becomes active with Router A as standby, given a similar priority setting. The two routers share IP traffic during normal operations. If one router is unavailable, the other router handles packet transfer functions.

Note Enter the **standby preempt** interface configuration command on the HSRP interfaces for MHSRP to ensure preemption restores load sharing after a router failure and reboot.

Figure 2: MHSRP load sharing



HSRP SSO

HSRP stateful switchover (SSO) alters the behavior of HSRP on devices configured for SSO redundancy mode with redundant Route Processors (RPs). When an RP is active and the other RP is standby, SSO enables the standby RP to take over if the active RP fails.

This functionality synchronizes HSRP SSO information to the standby RP, allowing continuous traffic forwarding using the HSRP virtual IP address during a switchover without data loss or path change. Additionally, if both RPs fail on the active HSRP device, the standby HSRP device becomes the active device.

The system automatically enables the feature when the redundancy mode is set to SSO.

HSRP and switch stacks

HSRP hello messages are generated by the active switch. If HSRP fails on the active switch, the HSRP active state may experience a flap. This is because HSRP hello messages are not generated while a new active switch is elected and initialized, and the standby router may become active after the active switch fails.

Default settings

Table 1: Default HSRP configuration

Feature	Default setting
HSRP version	Version 1
HSRP groups	None configured
Standby group number	0
Standby MAC address	System assigned as: 0000.0c07.acXX, where XX is the HSRP group number
Standby priority	100
Standby delay	0 (no delay)
Standby track interface priority	10
Standby hello time	3 seconds
Standby holdtime	10 seconds
HSRP IPv6 UDP port number	2029

HSRP configuration guidelines

Refer to these guidelines before configuring HSRP:

- HSRPv2 and HSRPv1 are mutually exclusive. HSRPv2 and HSRPv1 cannot interoperate on the same interface.
- In the procedures, the specified interface must be one of these Layer 3 interfaces:
  - Routed port: A physical port configured as a Layer 3 port by entering the **no switchport** command in interface configuration mode.
  - SVI: A VLAN interface created by using the **interface vlan** *vlan_id* in global configuration mode, and by default a Layer 3 interface.
  - Etherchannel port channel in Layer 3 mode: A port-channel logical interface created by using the **interface port-channel** *port-channel-number* in global configuration mode, and binding the Ethernet interface into the channel group.
- Assign IP addresses to all Layer 3 interfaces.

When configuring HSRP priority, follow these guidelines:

- Specify a priority to choose the active and standby routers. If preemption is enabled, the router with the highest priority becomes the active router. If priorities are equal, the current active router does not change.

- The highest number (1 to 255) represents the highest priority (most likely to become the active router).
- When setting the priority, preempt, or both, you must specify at least one keyword: **priority**, **preempt**, or both.
- Configure a delay time to allow the router to update its routing table when routing is first enabled. This ensures the router can provide adequate service and solve potential preemption issues.

When configuring HSRP authentication and timers, follow these guidelines:

- HSRP messages send the authentication string unencrypted. You must configure the same authentication string on all routers and access servers on a cable to ensure interoperation. An authentication mismatch prevents a device from learning the designated Hot Standby IP address and timer values from other routers configured with HSRP.
- Routers or access servers on which standby timer values are not configured can learn timer values from the active or standby router. The timers configured on an active router always override any other timer settings.
- All routers in a Hot Standby group should use the same timer values. Normally, the *holdtime* is greater than or equal to 3 times the *hellotime*.

Configure HSRP

This section provides configuration information about HSRP.

Enable HSRP

The **standby ip** interface configuration command activates HSRP on the configured interface. If an IP address is specified, that address is used as the designated address for the Hot Standby group. If no IP address is specified, the address is learned through the standby function. Configure at least one Layer 3 port on the LAN with the designated address. Configuring an IP address overrides the current designated address.

When the **standby ip** command is enabled on an interface and proxy ARP is enabled, if the interface's Hot Standby state is active, proxy ARP requests are answered using the Hot Standby group MAC address. Proxy ARP responses are suppressed if the interface is in a different state.

Procedure

	Command or Action	Purpose
Step 1	**configure terminal** **Example:** `Device(config)# configure terminal`	Enters global configuration mode.

	Command or Action	Purpose
Step 2	**interface** *interface-id* **Example:** `Device(config)# interface gigabitethernet1/0/1`	Enters interface configuration mode, and enter the Layer 3 interface on which you want to enable HSRP.
Step 3	**standby version** {**1** | **2**} **Example:** `Device(config-if)# standby version 1`	(Optional) Configures the HSRP version on the interface. • **1**: Selects HSRPv1. • **2**: Selects HSRPv2. Without a specified keyword, the interface defaults to HSRP v1.
Step 4	**standby** *group-number* **ip** [*ip-address* [**secondary**]] **Example:** `Device(config-if)# standby 1 ip`	Creates (or enable) the HSRP group using its number and virtual IP address. • *group-number*: The group number on the interface for which HSRP is being enabled. The range is 0 to 255; the default is 0. If there is only one HSRP group, you do not need to enter a group number. • (Optional on all but one interface) *ip-address*: The virtual IP address of the hot standby router interface. You must enter the virtual IP address for at least one of the interfaces; it can be learned on the other interfaces. • (Optional) **secondary**: The IP address is a secondary hot standby router interface. If neither router is designated as a secondary or standby router, and no priorities are set, the primary IP addresses are compared. The router with the higher IP address becomes the active router, and the router with the next highest IP address becomes the standby router.
Step 5	**end** **Example:** `Device(config-if)# end`	Returns to privileged EXEC mode
Step 6	**show standby** [*interface-id* [*group*]] **Example:** `Device# show standby`	Verifies the configuration of the standby groups.

	Command or Action	Purpose
Step 7	**copy running-config startup-config** **Example:** `Device# copy running-config startup-config`	(Optional) Saves your entries in the configuration file.

Enable an HSRP group for IPv6 operation

To enable an HSRP group for IPv6, perform this procedure:

Procedure

	Command or Action	Purpose
Step 1	**enable** **Example:** `Device> enable`	Enables privileged EXEC mode. • Enter your password if prompted.
Step 2	**configure terminal** **Example:** `Device# configure terminal`	Enters global configuration mode.
Step 3	**ipv6 unicast-routing** **Example:** `Device(config)# ipv6 unicast-routing`	Enables the forwarding of IPv6 unicast datagrams. • The **ipv6 unicast-routing** command must be enabled for HSRP for IPv6 to work.
Step 4	**interface** *type number* **Example:** `Device(config)# interface GigabitEthernet 0/0/0`	Specifies an interface type and number, and places the device in interface configuration mode.
Step 5	**standby** *group-number* **ipv6** {*link-local-address* | **autoconfig**} **Example:** `Device(config-if)# standby 1 ipv6 autoconfig`	Activates the HSRP in IPv6.
Step 6	**standby** *group-number* **preempt** [**delay** {**minimum** *seconds* | **reload** *seconds* | **sync** *seconds*}] **Example:**	Configures HSRP preemption and preemption delay.

	Command or Action	Purpose
	`Device(config-if)# standby 1 preempt`	
Step 7	**standby** *group-number* **priority** *priority* **Example:** `Device(config-if)# standby 1 priority 110`	Configures HSRP priority.
Step 8	**exit** **Example:** `Device(config-if)# exit`	Returns the device to privileged EXEC mode.
Step 9	**show standby** [*type number* [*group* | **summary**]] **Example:** `Device# show standby`	Displays HSRP information.
Step 10	**show ipv6 interface** *type number* **Example:** `Device# show ipv6 interface GigabitEthernet 0/0/0`	Displays the usability status of interfaces configured for IPv6.

Configure HSRP priority

The **standby priority**, **standby preempt**, and **standby track** interface configuration commands are all used to set characteristics for finding active and standby routers and behavior regarding when a new active router takes over.

Beginning in privileged EXEC mode, use one or more of these steps to configure HSRP priority characteristics on an interface:

Procedure

	Command or Action	Purpose
Step 1	**configure terminal** **Example:** `Device# configure terminal`	Enters global configuration mode.
Step 2	**interface** *interface-id* **Example:** `Device(config)# interface gigabitethernet1/0/1`	Enters interface configuration mode, and enter the HSRP interface on which you want to set priority.

	Command or Action	Purpose
Step 3	**standby** *group-number* **priority** *priority* **Example:** `Device(config-if)# ` **`standby 120 priority 50`**	Sets a **priority** value used in choosing the active router. The range is from 1 to 255; the default priority is 100. The highest number represents the highest priority. *group-number*: The group number to which the command applies. Use the **no** form of the command to restore the default values.
Step 4	**standby** *group-number* **preempt** [**delay** {**minimum** *seconds* | **reload** *seconds* | **sync** *seconds*}] **Example:** `Device(config-if)# ` **`standby 1 preempt delay minimum 300`**	Configures the router to **preempt**, which means that when the local router has a higher priority than the active router, it becomes the active router. • *group-number*: The group number to which the command applies. • (Optional) **delay minimum**: Set to cause the local router to postpone taking over the active role for the number of seconds shown. The range is from 0 to 3600 seconds (1 hour); the default is 0 (no delay before taking over). • (Optional) **delay reload**: Set to cause the local router to postpone taking over the active role after a reload by the indicated number of seconds. The range is from 0 to 3600 seconds (1 hour); the default is 0 (no delay before taking over after a reload). • (Optional) **delay sync**: Set to cause the local router to postpone taking over the active role so that IP redundancy clients can reply (either with an ok or wait reply) for the number of seconds shown. The range is from 0 to 3600 seconds (1 hour); the default is 0 (no delay before taking over). Use the **no** form of the command to restore the default values.
Step 5	**standby** *group-number* **track** *number* [**decrement** *decrement-value*] **Example:** `Device(config-if)# ` **`standby 1 track 1 decrement 20`**	Configures an interface to perform object priority tracking. • *group-number*: The group number to which the command applies.

	Command or Action	Purpose
		• *number*: Enter the object number (combined with interface type) that is tracked. • (Optional) **decrement** *decrement-value*: Enter the decrement value.
Step 6	**end** **Example:** `Device(config-if)# end`	Returns to privileged EXEC mode.
Step 7	**show running-config**	Verifies the configuration of the standby groups.
Step 8	**copy running-config startup-config**	(Optional) Saves your entries in the configuration file.

Configure MHSRP

To enable MHSRP and load-balancing, configure two routers as active routers for your groups, with virtual routers as standby routers, as shown in the *MHSRP Load Sharing* figure in the Multiple HSRP section. Enter the **standby preempt** interface configuration command on each HSRP interface. This ensures that if a router fails and comes back up, the preemption occurs and restores load-balancing.

Router A is configured as the active router for group 1, while Router B is configured as the active router for group 2. The HSRP interface for Router A has an IP address of 10.0.0.1 with a group 1 standby priority of 110 (the default is 100). The HSRP interface for Router B has an IP address of 10.0.0.2 with a group 2 standby priority of 110.

Group 1 uses a virtual IP address of 10.0.0.3 and group 2 uses a virtual IP address of 10.0.0.4.

Configure Router A

Procedure

	Command or Action	Purpose
Step 1	**configure terminal** **Example:** `Device# configure terminal`	Enters global configuration mode.
Step 2	**interface** *type number* **Example:** `Device(config)# interface gigabitethernet1/0/1`	Configures an interface type and enters interface configuration mode.

	Command or Action	Purpose
Step 3	**ip address** *ip-address mask* **Example:** `Device(config-if)# ip address 10.0.0.1 255.255.255.0`	Specifies an IP address for an interface.
Step 4	**standby** *group-number* **ip** [*ip-address* [**secondary**]] **Example:** `Device(config-if)# standby 1 ip 10.0.0.3`	Creates (or enable) the HSRP group using its number and virtual IP address. • *group-number*: The group number on the interface for which HSRP is being enabled. The range is 0 to 255; the default is 0. If there is only one HSRP group, you do not need to enter a group number. • (Optional on all but one interface) *ip-address*: The virtual IP address of the hot standby router interface. You must enter the virtual IP address for at least one of the interfaces; it can be learned on the other interfaces. • (Optional) **secondary**: The IP address is a secondary hot standby router interface. If neither router is designated as a secondary or standby router and no priorities are set, the primary IP addresses are compared and the higher IP address is the active router, with the next highest as the standby router.
Step 5	**standby** *group-number* **priority** *priority* **Example:** `Device(config-if)# standby 1 priority 110`	Sets a **priority** value used in choosing the active router. The range is 1 to 255; the default priority is 100. The highest number represents the highest priority. *group-number*: The group number to which the command applies. Use the **no** form of the command to restore the default values.
Step 6	**standby** *group-number* **preempt** [**delay** {**minimum** *seconds* | **reload** *seconds* | **sync** *seconds*}] **Example:** `Device(config-if)# standby 1 preempt delay minimum 300`	Configures the router to **preempt**, which means that when the local router has a higher priority than the active router, it becomes the active router. • *group-number*: The group number to which the command applies. • (Optional) **delay minimum**: Set to cause the local router to postpone taking over

	Command or Action	Purpose
		the active role for the number of seconds shown. The range is from 0 to 3600 seconds (1 hour); the default is 0 (no delay before taking over). • (Optional) **delay reload**: Set to cause the local router to postpone taking over the active role after a reload by the indicated number of seconds. The range is from 0 to 3600 seconds (1 hour); the default is 0 (no delay before taking over after a reload). • (Optional) **delay sync**: Set to cause the local router to postpone taking over the active role so that IP redundancy clients can reply (either with an ok or wait reply) for the number of seconds shown. The range is from 0 to 3600 seconds (1 hour); the default is 0 (no delay before taking over). Use the **no** form of the command to restore the default values.
Step 7	**standby** *group-number* **ip** [*ip-address* [**secondary**]] **Example:** `Device(config-if)# standby 1 ip 10.0.0.4`	Creates (or enable) the HSRP group using its number and virtual IP address. • *group-number*: The group number on the interface for which HSRP is being enabled. The range is 0 to 255; the default is 0. If there is only one HSRP group, you do not need to enter a group number. • (Optional on all but one interface) *ip-address*: The virtual IP address of the hot standby router interface. You must enter the virtual IP address for at least one of the interfaces; it can be learned on the other interfaces. • (Optional) **secondary**: The IP address is a secondary hot standby router interface. If neither router is designated as a secondary or standby router and no priorities are set, the primary IP addresses are compared and the higher IP address is the active router, with the next highest as the standby router.

	Command or Action	Purpose
Step 8	**standby** *group-number* **preempt** [**delay** {**minimum** *seconds* | **reload** *seconds* | **sync** *seconds*}] **Example:** `Device(config-if)# standby 2 preempt delay minimum 300`	Configures the router to **preempt**, which means that when the local router has a higher priority than the active router, it becomes the active router. • *group-number*: The group number to which the command applies. • (Optional) **delay minimum**: Set to cause the local router to postpone taking over the active role for the number of seconds shown. The range is from 0 to 3600 seconds (1 hour); the default is 0 (no delay before taking over). • (Optional) **delay reload**: Set to cause the local router to postpone taking over the active role after a reload by the indicated number of seconds. The range is from 0 to 3600 seconds (1 hour); the default is 0 (no delay before taking over after a reload). • (Optional) **delay sync**: Set to cause the local router to postpone taking over the active role so that IP redundancy clients can reply (either with an ok or wait reply) for the number of seconds shown. The range is from 0 to 3600 seconds (1 hour); the default is 0 (no delay before taking over). Use the **no** form of the command to restore the default values.
Step 9	**end** **Example:** `Device(config-if)# end`	Returns to privileged EXEC mode.
Step 10	**show running-config**	Verifies the configuration of the standby groups.
Step 11	**copy running-config startup-config**	(Optional) Saves your entries in the configuration file.

Configure Router B

Procedure

	Command or Action	Purpose
Step 1	**configure terminal** **Example:** `Device# configure terminal`	Enters global configuration mode.
Step 2	**interface** *type number* **Example:** `Device(config)# interface gigabitethernet1/0/1`	Configures an interface type and enters interface configuration mode.
Step 3	**ip address** *ip-address mask* **Example:** `Device(config-if)# ip address 10.0.0.2 255.255.255.0`	Specifies an IP address for an interface.
Step 4	**standby** *group-number* **ip** [*ip-address* [**secondary**]] **Example:** `Device(config-if)# standby 1 ip 10.0.0.3`	Creates (or enable) the HSRP group using its number and virtual IP address. • *group-number*: The group number on the interface for which HSRP is being enabled. The range is 0 to 255; the default is 0. If there is only one HSRP group, you do not need to enter a group number. • (Optional on all but one interface) *ip-address*: The virtual IP address of the hot standby router interface. You must enter the virtual IP address for at least one of the interfaces; it can be learned on the other interfaces. • (Optional) **secondary**: The IP address is a secondary hot standby router interface. If neither router is designated as a secondary or standby router, and no priorities are set, the primary IP addresses are compared. The router with the higher IP address becomes the active router, and the router with the next highest IP address becomes the standby router.
Step 5	**standby** *group-number* **priority** *priority* **Example:**	Sets a **priority** value used in choosing the active router. The range is from 1 to 255; the

	Command or Action	Purpose
	`Device(config-if)# standby 2 priority 110`	default priority is 100. The highest number represents the highest priority. *group-number*: The group number to which the command applies. Use the **no** form of the command to restore the default values.
Step 6	**standby** *group-number* **preempt** [**delay** {**minimum** *seconds* | **reload** *seconds* | **sync** *seconds*}] **Example:** `Device(config-if)# standby 1 preempt delay minimum 300`	Configures the router to **preempt**, which means that when the local router has a higher priority than the active router, it becomes the active router. • *group-number*: The group number to which the command applies. • (Optional) **delay minimum**: Set to cause the local router to postpone taking over the active role for the number of seconds shown. The range is from 0 to 3600 seconds (1 hour); the default is 0 (no delay before taking over). • (Optional) **delay reload**: Set to cause the local router to postpone taking over the active role after a reload by the indicated number of seconds. The range is from 0 to 3600 seconds (1 hour); the default is 0 (no delay before taking over after a reload). • (Optional) **delay sync**: Set to cause the local router to postpone taking over the active role so that IP redundancy clients can reply (either with an ok or wait reply) for the number of seconds shown. The range is from 0 to 3600 seconds (1 hour); the default is 0 (no delay before taking over). Use the **no** form of the command to restore the default values.
Step 7	**standby** *group-number* **ip** [*ip-address* [**secondary**]] **Example:** `Device(config-if)# standby 2 ip 10.0.0.4`	Creates the HSRP group using its number and virtual IP address. • *group-number*: The group number on the interface for which HSRP is being enabled. The range is 0 to 255; the default is 0. If there is only one HSRP group, you do not need to enter a group number.

	Command or Action	Purpose
		• (Optional on all but one interface) *ip-address*: The virtual IP address of the hot standby router interface. You must enter the virtual IP address for at least one of the interfaces; it can be learned on the other interfaces. • (Optional) **secondary**: The IP address is a secondary hot standby router interface. If neither router is designated as a secondary or standby router and no priorities are set, the primary IP addresses are compared and the higher IP address is the active router, with the next highest as the standby router.
Step 8	**standby** *group-number* **preempt** [**delay** {**minimum** *seconds* | **reload** *seconds* | **sync** *seconds*}] **Example:** `Device(config-if)# standby 2 preempt delay minimum 300`	Configures the router to **preempt**, which means that when the local router has a higher priority than the active router, it becomes the active router. • *group-number*: The group number to which the command applies. • (Optional) **delay minimum**: Set to cause the local router to postpone taking over the active role for the number of seconds shown. The range is from 0 to 3600 seconds (1 hour); the default is 0 (no delay before taking over). • (Optional) **delay reload**: Set to cause the local router to postpone taking over the active role after a reload by the indicated number of seconds. The range is from 0 to 3600 seconds (1 hour); the default is 0 (no delay before taking over after a reload). • (Optional) **delay sync**: Set to cause the local router to postpone taking over the active role so that IP redundancy clients can reply (either with an ok or wait reply) for the number of seconds shown. The range is from 0 to 3600 seconds (1 hour); the default is 0 (no delay before taking over). Use the **no** form of the command to restore the default values.

	Command or Action	Purpose
Step 9	**end** **Example:** `Device(config-if)# end`	Returns to privileged EXEC mode.
Step 10	**show running-config**	Verifies the configuration of the standby groups.
Step 11	**copy running-config startup-config**	(Optional) Saves your entries in the configuration file.

Configure HSRP authentication and timers

You can optionally configure an HSRP authentication string or change the hello-time interval and hold-time interval.

Beginning in privileged EXEC mode, use one or more of these steps to configure HSRP authentication and timers on an interface:

Procedure

	Command or Action	Purpose
Step 1	**configure terminal** **Example:** `Device# configure terminal`	Enters global configuration mode.
Step 2	**interface** *interface-id* **Example:** `Device(config)# interface gigabitethernet1/0/1`	Enters interface configuration mode, and enter the HSRP interface on which you want to set priority.
Step 3	**standby** *group-number* **authentication** *string* **Example:** `Device(config-if)# standby 1 authentication word`	**authentication** *string*: Enter a string to be carried in all HSRP messages. The authentication string can be up to eight characters in length; the default string is **cisco**. *group-number*: The group number to which the command applies.
Step 4	**standby** *group-number* **timers** *hellotime holdtime* **Example:** `Device(config-if)# standby 1 timers 5 15`	Configures the time interval to send and receive hello packets. • *group-number*: The group number to which the command applies.

	Command or Action	Purpose
		• *hellotime*: Set the interval between successive hello packets in seconds. The range is 1 to 254 seconds. • *holdtime*: Set the interval to wait for a hello packet from a neighbor device before declaring the neighbor device as inactive. The range is 34 to 255 seconds.
Step 5	**end** **Example:** `Device(config-if)# end`	Returns to privileged EXEC mode.
Step 6	**show running-config**	Verifies the configuration of the standby groups.
Step 7	**copy running-config startup-config**	(Optional) Saves your entries in the configuration file.

Configure HSRP groups and clustering

When your device participates in HSRP standby routing and clustering is enabled, use the same standby group for command switch redundancy and HSRP redundancy.

Procedure

	Command or Action	Purpose
Step 1	**configure terminal** **Example:** `Device# configure terminal`	Enters global configuration mode.
Step 2	**cluster standby-group** *HSRP-group-name* [**routing-redundancy**] **Example:** `Device(config)# cluster standby-group my_hsrp routing-redundancy`	Binds the HSRP standby group to the cluster and enables the same HSRP standby group to be used for command switch and routing redundancy. HSRP standby routing is disabled for the group if you create a cluster with the same HSRP standby group name without entering the **routing-redundancy** keyword.
Step 3	**end** **Example:** `Device(config)# end`	Returns to privileged EXEC mode.

Configuration examples

The following sections provide various configuration examples for HSRP.

Example: Enable HSRP

This example shows how to activate HSRP for group 1 on an interface. The IP address used by the hot standby group is learned by using HSRP.

Note This procedure is the minimum number of steps required to enable HSRP. Other configurations are optional.

```
Device# configure terminal
Device(config)# interface gigabitethernet1/0/1
Device(config-if)# no switchport
Device(config-if)# standby 1 ip
Device(config-if)# end
Device# show standby
```

Example: Configure and verify an HSRP group

These examples show configuration and verification for an HSRP group for IPv6 that consists of Device 1 and Device 2.

You can display HSRP information for the entire switch, a specific interface, an HSRP group, or an HSRP group on an interface. You can also specify whether to display a concise overview of HSRP information or detailed HSRP information. The default display is **detail**. If there are a large number of HSRP groups, using the **show standby** command without qualifiers can result in an unwieldy display.

Device 1 configuration

```
interface FastEthernet0/0.100
description DATA VLAN for PCs
encapsulation dot1Q 100
ipv6 address 2001:DB8:CAFE:2100::BAD1:1010/64
standby version 2
standby 101 priority 120
standby 101 preempt delay minimum 30
standby 101 authentication ese
standby 101 track 1 decrement 20
standby 201 ipv6 autoconfig
standby 201 priority 120
standby 201 preempt delay minimum 30
standby 201 authentication ese
standby 201 track 1 decrement 20

Device1# show standby

FastEthernet0/0.100 - Group 101 (version 2)
State is Active
2 state changes, last state change 5w5d
```

```
Active virtual MAC address is 0000.0c9f.f065
Local virtual MAC address is 0000.0c9f.f065 (v2 default)
Hello time 3 sec, hold time 10 sec
Next hello sent in 2.296 secs
Authentication text "ese"
Preemption enabled, delay min 30 secs
Active router is local
Priority 120 (configured 120)
Track 1 state Up decrement 20
IP redundancy name is "hsrp-Fa0/0.100-101" (default)
FastEthernet0/0.100 - Group 201 (version 2)
State is Active
2 state changes, last state change 5w5d
Virtual IP address is FE80::5:73FF:FEA0:C9
Active virtual MAC address is 0005.73a0.00c9
Local virtual MAC address is 0005.73a0.00c9 (v2 IPv6 default)
Hello time 3 sec, hold time 10 sec
Next hello sent in 2.428 secs
Authentication text "ese"
Preemption enabled, delay min 30 secs
Active router is local
Standby router is FE80::20F:8FFF:FE37:3B70, priority 100 (expires in 7.856 sec)
Priority 120 (configured 120)
Track 1 state Up decrement 20
IP redundancy name is "hsrp-Fa0/0.100-201" (default)
```

Device 2 configuration

```
interface FastEthernet0/0.100
description DATA VLAN for Computers
encapsulation dot1Q 100
ipv6 address 2001:DB8:CAFE:2100::BAD1:1020/64
standby version 2
standby 101 preempt
standby 101 authentication ese
standby 201 ipv6 autoconfig
standby 201 preempt
standby 201 authentication ese
```

```
Device2# show standby

FastEthernet0/0.100 - Group 101 (version 2)
State is Standby
7 state changes, last state change 5w5d
Active virtual MAC address is 0000.0c9f.f065
Local virtual MAC address is 0000.0c9f.f065 (v2 default)
Hello time 3 sec, hold time 10 sec
Next hello sent in 0.936 secs
Authentication text "ese"
Preemption enabled
MAC address is 0012.7fc6.8f0c
Standby router is local
Priority 100 (default 100)
IP redundancy name is "hsrp-Fa0/0.100-101" (default)
FastEthernet0/0.100 - Group 201 (version 2)
State is Standby
7 state changes, last state change 5w5d
Virtual IP address is FE80::5:73FF:FEA0:C9
Active virtual MAC address is 0005.73a0.00c9
Local virtual MAC address is 0005.73a0.00c9 (v2 IPv6 default)
Hello time 3 sec, hold time 10 sec
Next hello sent in 0.936 secs
Authentication text "ese"
Preemption enabled
Active router is FE80::212:7FFF:FEC6:8F0C, priority 120 (expires in 7.548 sec)
```

```
MAC address is 0012.7fc6.8f0c
Standby router is local
Priority 100 (default 100)
IP redundancy name is "hsrp-Fa0/0.100-201" (default)
```

Example: Configure HSRP priority

This example activates a port, sets an IP address and a priority of 120 (higher than the default value), and waits for 300 seconds (5 minutes) before attempting to become the active router:

```
Device# configure terminal
Device(config)# interface gigabitethernet1/0/1
Device(config-if)# standby ip 172.20.128.3
Device(config-if)# standby priority 120 preempt delay minimum 300
Device(config-if)# end
Device# show standby
```

Example: Configure MHSRP

This example shows how to enable the MHSRP configuration shown in the figure *MHSRP load sharing*.

Router A configuration

```
Device# configure terminal
Device(config)# interface gigabitethernet1/0/1
Device(config-if)# ip address 10.0.0.1 255.255.255.0
Device(config-if)# standby ip 10.0.0.3
Device(config-if)# standby 1 priority 110
Device(config-if)# standby 1 preempt
Device(config-if)# standby 2 ip 10.0.0.4
Device(config-if)# standby 2 preempt
Device(config-if)# end
```

Router B configuration

```
Device# configure terminal
Device(config)# interface gigabitethernet1/0/1
Device(config-if)# ip address 10.0.0.2 255.255.255.0
Device(config-if)# standby ip 10.0.0.3
Device(config-if)# standby 1 preempt
Device(config-if)# standby 2 ip 10.0.0.4
Device(config-if)# standby 2 priority 110
Device(config-if)# standby 2 preempt
Device(config-if)# end
```

Example: Configure HSRP authentication and timer

This example shows how to configure word as the authentication string required to allow Hot Standby routers in group 1 to interoperate:

```
Device# configure terminal
Device(config)# interface gigabitethernet1/0/1
```

```
Device(config-if)# standby 1 authentication word
Device(config-if)# end
```

This example shows how to set the timers on standby group 1 with the time between hello packets at 5 seconds and the time after which a router is considered down to be 15 seconds:

```
Device# configure terminal
Device(config)# interface gigabitethernet1/0/1
Device(config-if)# standby 1 ip
Device(config-if)# standby 1 timers 5 15
Device(config-if)# end
```

Example: Configure HSRP groups and clustering

This example shows how to bind standby group my_hsrp to the cluster and enable the same HSRP group to be used for command switch redundancy and router redundancy. The command can only be executed on the cluster command switch. If the standby group name or number does not exist, or if the switch is a cluster member switch, an error message appears.

```
Device# configure terminal
Device(config)# cluster standby-group my_hsrp routing-redundancy
Device(config-if)# end
```

CHAPTER 2

VRRP

Feature history for VRRP

This table provides release and platform support information for the features explained in this module.

These features are available in all the releases subsequent to the one they were introduced in, unless noted otherwise.

Release	Feature name and description	Supported platform
Cisco IOS XE 26.2.1ea	VRRP: VRRP feature support has been introduced.	Cisco C9550 Series Smart Switches
Cisco IOS XE 17.18.1	VRRP: VRRP is an election protocol that dynamically assigns responsibility for one or more virtual devices to the VRRP devices on a LAN, allowing several devices on a multi-access link to utilize the same virtual IP address.	Cisco C9350 Series Smart Switches Cisco C9610 Series Smart Switches

Understand VRRP

The Virtual Router Redundancy Protocol (VRRP) is an election protocol that dynamically assigns responsibility for one or more virtual devices to the VRRP devices on a LAN, allowing several devices on a multiaccess link to utilize the same virtual IP address. You configure a VRRP device to run the VRRP protocol with other devices on a LAN. In a VRRP configuration, one device is elected as the virtual primary device. The other devices act as backups if the virtual primary device fails.

VRRP benefits

- Redundancy: Enables you to configure multiple devices as the default gateway device, which reduces the possibility of a single point of failure in a network.
- Load sharing: Allows traffic to and from LAN clients to be shared by multiple devices. The traffic load is shared more equitably among available devices.
- Multiple VRRP groups: Supports up to 255 virtual devices (VRRP groups) on a device physical interface, subject to restrictions in scaling. Multiple VRRP groups enable you to implement redundancy and load sharing in your LAN topology.
- Multiple IP addresses: Allows you to manage multiple IP addresses, including secondary IP addresses. If you have multiple subnets that are configured on an Ethernet interface, you can configure VRRP on each subnet.
- IPv4 and IPv6: VRRPv3 supports IPv4 and IPv6 address families while VRRPv2 only supports IPv4 addresses.
- Preemption: Enables you to preempt a backup device that has taken over for a failing primary with a higher priority backup device that has become available.
- Advertisement protocol: Uses a dedicated Internet Assigned Numbers Authority (IANA) standard multicast address (224.0.0.18) for VRRP advertisements. This addressing scheme minimizes the number of devices that must service the multicasts and allows test equipment to accurately identify VRRP packets on a segment. IANA has assigned the IP protocol number 112 to VRRP.
- VRRP object tracking: Ensures that the best VRRP device is the primary for the group by altering VRRP priorities based on interface states.
- SSO: VRRPv3 supports Stateful Switchover (SSO). Enable the **fhrp sso** command for VRRPv3 to support SSO. Disable SSO support using the **no fhrp sso** command.

VRRP operation

There are several ways a LAN client can determine which device should be the first hop to a particular remote destination. The client can use a dynamic process or static configuration. Examples of dynamic device discovery are as follows:

- Proxy Address Resolution Protocol (ARP): The client uses ARP to get the destination it wants to reach, and a device responds to the ARP request with its own MAC address.
- Routing protocol: The client listens to updates from dynamic routing protocols like Routing Information Protocol (RIP) and then forms its own routing table.
- ICMP Router Discovery Protocol (IRDP): The client runs an Internet Control Message Protocol (ICMP) device discovery client.

Dynamic discovery protocols have the drawback of requiring configuration and processing overhead on your LAN client. In addition, in the event of a device failure, the process of switching to another device can be slow.

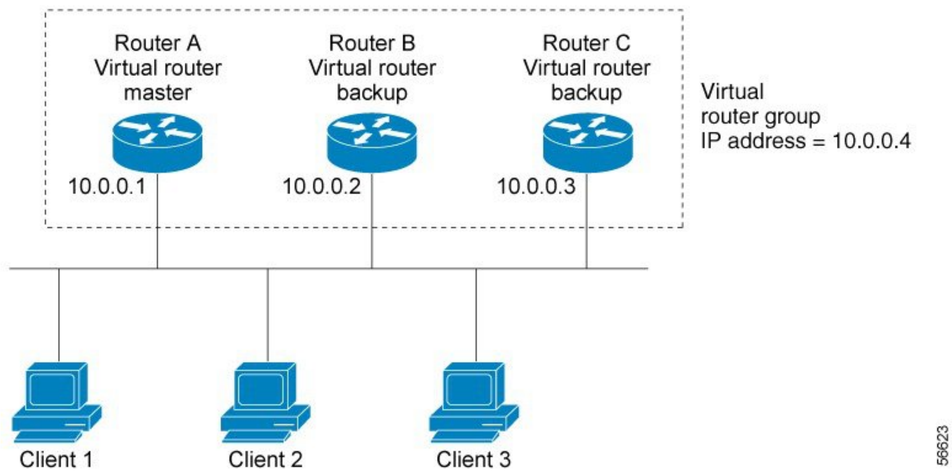
Statically configuring a default device on the client is an alternative to dynamic discovery protocols. This approach simplifies client configuration and processing, but creates a single point of failure. If the default gateway fails, the LAN client is limited to communicating only on the local IP network segment and is cut off from the rest of the network.

VRRP solves the static configuration problem by enabling a group of devices to form a single virtual device. The LAN clients can then be configured with the virtual device as their default gateway. The virtual device, representing a group of devices, is also known as a VRRP group.

VRRP is supported on several interfaces including Ethernet, Fast Ethernet, BVI, and Gigabit Ethernet. It also extends support to MPLS VPNs, VRF-aware MPLS VPNs, and VLANs.

A LAN topology configured with VRRP is illustrated by this figure. In this example, Routers A, B, and C run VRRP and form a virtual device. The IP address of the virtual device matches that configured for Router A's Ethernet interface, 10.0.0.1.

Figure 3: Basic VRRP topology

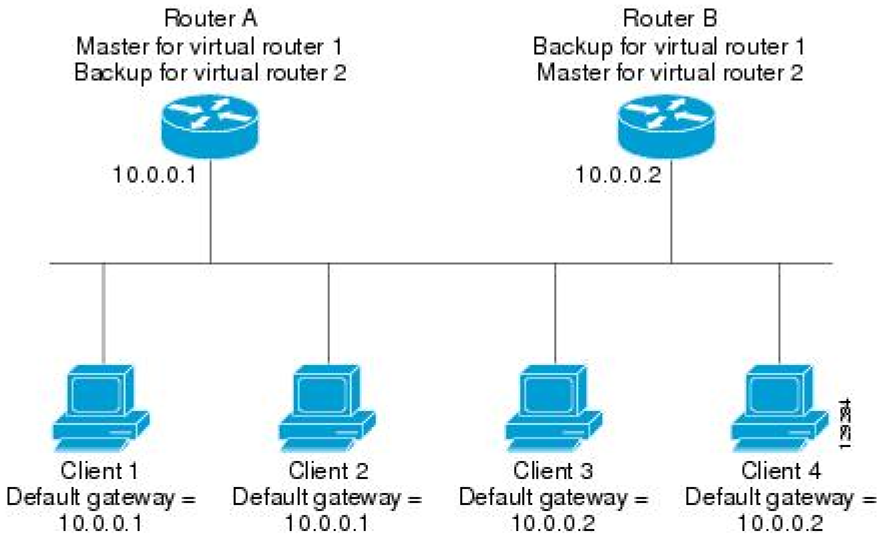


Because the virtual device uses the IP address of the physical Ethernet interface of Router A, Router A assumes the role of the virtual primary device and is also known as the IP address owner. As the virtual primary device, Router A controls the IP address of the virtual device and is responsible for forwarding packets sent to this IP address. Clients 1 to 3 are configured with the default gateway IP address of 10.0.0.1.

Routers B and C serve as backups for the virtual device. If the virtual primary device fails, the device configured with the higher priority will become the virtual primary device and provide uninterrupted service for the LAN hosts. When Router A recovers, it becomes the virtual primary device again. For more detail on the roles of VRRP devices and the process when the virtual primary device fails, see the VRRP device priority and preemption, on page 28 section.

The figure below shows a LAN topology in which VRRP is configured so that Routers A and B share the traffic to and from clients 1 through 4 and that Routers A and B act as virtual device backups to each other if either device fails.

Figure 4: Load sharing and redundancy VRRP topology



In this topology, two virtual devices are configured. For virtual device 1, Router A is the owner of IP address 10.0.0.1 and virtual primary device, and Router B is the virtual device backup to Router A. Clients 1 and 2 are configured with the default gateway IP address of 10.0.0.1.

For virtual device 2, Router B is the owner of IP address 10.0.0.2 and virtual primary device, and Router A is the virtual device backup to Router B. Clients 3 and 4 are configured with the default gateway IP address of 10.0.0.2.

VRRP device priority and preemption

VRRP device priority is an important aspect of the VRRP redundancy scheme. Priority determines each VRRP device's role and the actions if the virtual primary device fails.

If a VRRP device owns the IP address of the virtual device and the IP address of the physical interface, this device will function as a virtual primary device.

Priority also determines if a VRRP device functions as a virtual device backup and the order of ascendancy to becoming virtual primary device if the virtual primary device fails. Configure each virtual device backup's priority from 1 to 254 using the **priority** command (use the **vrrp address-family** command to enter the VRRP configuration mode and access the **priority** option).

For example, if Device A, the virtual primary device in a LAN topology, fails, an election process takes place to determine if virtual device backups B or C should take over. If Devices B and C are configured with the priorities of 101 and 100, respectively, Device B is elected to become virtual primary device because it has the higher priority. If Devices B and C are configured with a priority of 100, the backup device with the higher IP address becomes the virtual primary device.

By default, a preemptive scheme allows a higher-priority backup device to take over when available, replacing the device that initially became the primary. You can disable this preemptive scheme using the **no preempt** command (use the **vrrp address-family** command to enter the VRRP configuration mode, and enter the **no preempt** command). If preemption is disabled, the virtual device backup that is elected to become virtual primary device remains as the primary until the original virtual primary device recovers and becomes the primary again.

VRRP advertisements

The primary virtual device sends VRRP advertisements to other VRRP devices in the same group. The advertisements communicate the priority and state of the primary virtual device. The VRRP advertisements are encapsulated into either IPv4 or IPv6 packets (based on the VRRP group configuration) and sent to the appropriate multicast address assigned to the VRRP group. For IPv4, the multicast address is 224.0.0.18. For IPv6, the multicast address is FF02:0:0:0:0:0:0:12. The advertisements are sent every second by default and the interval is configurable.

Cisco devices allow you to configure millisecond timers, representing a change from VRRPv2. You need to manually configure the millisecond timer values on both the primary and the backup devices. The **show vrrp** command output on backup devices displays the primary advertisement value as 1 second; packets do not accept millisecond values.

Use millisecond timers only when necessary, with careful consideration and testing. Millisecond values function correctly under specific conditions. The millisecond timer values are compatible with other vendor equipment, as long as it supports VRRPv3. You can specify a timer value between 100 milliseconds and 40000 milliseconds.

VRRP object tracking

Object tracking is an independent process used to create, monitor, and remove tracked objects, such as the state of the line protocol of an interface. Clients such as the Hot Standby Router Protocol (HSRP), Gateway Load Balancing Protocol (GLBP), and VRRP register their interests with specific tracked objects and act when the states of objects change.

Each tracked object is identified by a unique number that is specified on the tracking CLI. Client processes such as VRRP use this number to track a specific object.

The tracking process periodically polls tracked objects and notes changes. The changes in the tracked object are communicated to interested client processes, either immediately or after a specified delay. The object values are reported as either up or down.

VRRP object tracking gives VRRP access to all the objects available through the tracking process. The tracking process allows you to track individual objects, such as the state of an interface line protocol, the state of an IP route, or the reachability of a route.

VRRP provides an interface to the tracking process. Each VRRP group can track multiple objects that may affect the priority of the VRRP device. You specify the object number to be tracked and VRRP is notified of any change to the object. VRRP increments (or decrements) the priority of the virtual device based on the state of the object being tracked.

How VRRP object tracking affects the priority of a device

The priority of a device can change dynamically with object tracking when the tracked object goes down. The tracking process periodically polls tracked objects and notes changes. The changes in the tracked object are communicated to VRRP, either immediately or after a specified delay. The object values are reported as either up or down.

You can track objects like the line protocol state of an interface or the reachability of an IP route. If the specified object goes down, the VRRP priority is reduced. A VRRP device with a higher priority can become the virtual primary device if you configure the **vrrp preempt** command.

VRRP In Service Software Upgrade

VRRP supports In Service Software Upgrade (ISSU), enabling a high-availability (HA) system to operate in stateful switchover (SSO) mode, even when different versions of Cisco IOS XE software run on the active and standby Route Processors (RPs) or line cards.

With ISSU, you can upgrade or downgrade between supported Cisco IOS XE releases without interrupting packet forwarding or sessions, thus reducing planned downtime. This is achieved by temporarily running different software versions on the active and standby RPs to maintain state information. This feature allows the system to continue forwarding packets without losing sessions and with minimal or no packet loss by switching to secondary RPs running upgraded (or downgraded) software. This feature is enabled by default.

VRRP Stateful Switchover

With the introduction of the VRRP Support for Stateful Switchover feature, VRRP is SSO aware. VRRP can detect when a device is failing over to the secondary RP and continue in its current group state.

SSO works in networking devices that usually serve as edge devices and support dual Route Processors (RPs). SSO provides RP redundancy by establishing one RP as active while the other remains in standby mode. SSO synchronizes critical state information between the RPs, ensuring the dynamic maintenance of network state information.

Before VRRP became SSO aware, deploying VRRP on a device with redundant RPs would cause the device to stop its VRRP group activities and rejoin the group as if it were reloaded when a switchover occurred. VRRP continues its activities as a group member during a switchover with the SSO-VRRP feature. VRRP state information is maintained between redundant RPs so that the standby RP can continue the device’s activities within the VRRP during and after a switchover.

This feature is enabled by default. Disable this feature using the **no vrrp sso** command in global configuration mode.

Restrictions for VRRP

- On Cisco C9610 Series Smart Switches, VRRP is not supported on subinterfaces.
- VRRP or VRRPv3 is designed for use over Ethernet LANs that are capable of multiaccess, multicast, or broadcast, and are not intended as a replacement for existing dynamic protocols.
- Do not configure the VRRP/VRRPv3 advertise timer to be less than the forwarding delay on the BVI interface. If you configure the VRRP/VRRPv3 advertise timer to a value equal to or greater than the forwarding delay on the BVI interface, the setting stops a VRRP/VRRPv3 device on a newly initialized BVI interface from taking over the primary role unconditionally.

  Use the **bridge forward-time** command to set the forwarding delay on the BVI interface. Use the **vrrp timers advertise** command to set the VRRP/VRRPv3 advertisement timer.
- Full network redundancy can only be achieved if VRRP operates over the same network path as the VRRS Pathway redundant interfaces. For full redundancy, these restrictions apply:
  - VRRS pathways should neither share a different physical interface as the parent VRRP group, nor be configured on a sub-interface having a different physical interface as the parent VRRP group.
  - VRRS pathways should not be configured on SVIs unless the associated VLANs share the same trunk as the VLAN for the parent VRRP group.

- The interface link-local IP address and the VRRP group virtual link-local IP address should be different for VRRP features to function properly.

Configure VRRP

This section provides information about the various tasks to configure VRRP.

Customize VRRP

Customizing the behavior of VRRP is optional. Be aware that enabling a VRRP group means the group is immediately operational. If you enable a VRRP group before customizing VRRP, the device might take control of the group and become the virtual primary device before customization is completed. Therefore, if you plan to customize VRRP, it is a good idea to do so before enabling VRRP.

Procedure

	Command or Action	Purpose
Step 1	**enable** **Example:** `Device> enable`	Enables privileged EXEC mode. Enter your password if prompted.
Step 2	**configure terminal** **Example:** `Device# configure terminal`	Enters global configuration mode.
Step 3	**interface** *type number* **Example:** `Device(config)# GigabitEthernet 0/0/0`	Enters interface configuration mode.
Step 4	**ip address** *ip-address mask* **Example:** `Device(config-if)# ip address 172.16.6.5 255.255.255.0`	Configures an IP address for an interface.
Step 5	**vrrp** *group* **description** *text* **Example:** `Device(config-if)# vrrp 10 description working-group`	Assigns a text description to the VRRP group.
Step 6	**vrrp** *group* **priority** *level* **Example:**	Sets the priority level of the device within a VRRP group.

	Command or Action	Purpose
	`Device(config-if)# vrrp 10 priority 110`	• The default priority is 100.
Step 7	**vrrp** *group* **preempt** [**delay minimum** *seconds*] **Example:** `Device(config-if)# vrrp 10 preempt delay minimum 380`	Configures the device to take over as virtual primary device for a VRRP group if it has a higher priority than the current virtual primary device. • The default delay period is 0 seconds.
Step 8	**vrrp** *group* **timers advertise** [**msec**] *interval* **Example:** `Device(config-if)# vrrp 10 timers advertise 110`	Configures the interval between successive advertisements by the virtual primary device in a VRRP group. • The unit of the interval is in seconds unless the **msec** keyword is specified. The default *interval* value is 1 second. **Note** All devices in a VRRP group must use the same timer values. If the same timer values are not set, the devices in the VRRP group will not communicate with each other and any misconfigured device will change its state to primary.
Step 9	**vrrp** *group* **timers learn** **Example:** `Device(config-if)# vrrp 10 timers learn`	Configures the device, when it is acting as virtual device backup for a VRRP group, to learn the advertisement interval used by the virtual primary device.
Step 10	**exit** **Example:** `Device(config-if)# exit`	Exits interface configuration mode.

Enable VRRP

To enable VRRP, perform this task.

Procedure

	Command or Action	Purpose
Step 1	**enable** **Example:** `Device> enable`	Enables privileged EXEC mode. Enter your password if prompted.

	Command or Action	Purpose
Step 2	**configure terminal** **Example:** `Device# configure terminal`	Enters global configuration mode.
Step 3	**interface** *type number* **Example:** `Device(config)# interface GigabitEthernet 0/0/0`	Enters interface configuration mode.
Step 4	**ip address** *ip-address mask* **Example:** `Device(config-if)# ip address 172.16.6.5 255.255.255.0`	Configures an IP address for an interface.
Step 5	**vrrp** *group* **ip** *ip-address* [**secondary**] **Example:** `Device(config-if)# vrrp 10 ip 172.16.6.1`	Enables VRRP on an interface. • Identify a primary IP address, and use the **vrrp ip** command with the **secondary** keyword to add additional IP addresses to the group. **Note** All devices in the VRRP group must be configured with the same primary address and a matching list of secondary addresses for the virtual device. Devices in the VRRP group will not communicate if different primary or secondary addresses are set, and any misconfigured device will change its state to primary.
Step 6	**end** **Example:** `Device(config-if)# end`	Returns to privileged EXEC mode.

Configure VRRP object tracking

Note The priority of a VRRP group, when it is the IP address owner, is set to a fixed value of 255, which object tracking cannot reduce.

Procedure

	Command or Action	Purpose
Step 1	**enable** **Example:** `Device> enable`	Enables privileged EXEC mode. Enter your password if prompted.
Step 2	**configure terminal** **Example:** `Device# configure terminal`	Enters global configuration mode.
Step 3	**track** *object-number* **interface** *type number* {**line-protocol** | **ip routing**} **Example:** `Device(config)# track 2 interface serial 6 line-protocol`	Configures an interface to be tracked where changes in the state of the interface affect the priority of a VRRP group. • This command configures the interface and corresponding object number to be used with the **vrrp track** command. • The **line-protocol** keyword tracks whether the interface is up. The **ip routing** keyword also checks that IP routing is enabled and active on the interface. • You can also use the **track ip route** command to track the reachability of an IP route or a metric type object.
Step 4	**interface** *type number* **Example:** `Device(config)# interface Ethernet 2`	Enters interface configuration mode.
Step 5	**vrrp** *group* **ip** *ip-address* **Example:** `Device(config-if)# vrrp 1 ip 10.0.1.20`	Enables VRRP on an interface and identifies the IP address of the virtual device.
Step 6	**vrrp** *group* **priority** *level* **Example:** `Device(config-if)# vrrp 1 priority 120`	Sets the priority level of the device within a VRRP group.
Step 7	**vrrp** *group* **track** *object-number* [**decrement** *priority*] **Example:**	Configures VRRP to track an object.

	Command or Action	Purpose
	`Device(config-if)# vrrp 1 track 2 decrement 15`	
Step 8	**end** **Example:** `Device(config-if)# end`	Returns to privileged EXEC mode.
Step 9	**show track** [*object-number*] **Example:** `Device# show track 1`	Displays tracking information.

Configure VRRP text authentication

Before you begin

The system does not enable interoperability with vendors who may have implemented the RFC 2338 method.

Procedure

	Command or Action	Purpose
Step 1	**enable** **Example:** `Device> enable`	Enables privileged EXEC mode. Enter your password if prompted.
Step 2	**configure terminal** **Example:** `Device# configure terminal`	Enters global configuration mode.
Step 3	**interface** *type number* **Example:** `Device(config)# interface GigabitEthernet 0/0/0`	Configures an interface type and enters interface configuration mode.
Step 4	**ip address** *ip-address mask* [**secondary**] **Example:** `Device(config-if)# ip address 10.0.0.1 255.255.255.0`	Specifies a primary or secondary IP address for an interface.
Step 5	**vrrp** *group* **authentication text** *text-string* **Example:**	Authenticates VRRP packets received from other devices in the group. • If you configure authentication, all devices within the VRRP group must use the same

	Command or Action	Purpose
	`Device(config-if)# vrrp 1 authentication text textstring1`	authentication string, which by default is 'cisco'. **Note** All devices within the VRRP group must be configured with the same authentication string. If inconsistent authentication strings are used, devices will fail to communicate, and any misconfigured device will change its state to primary.
Step 6	**vrrp** *group* **ip** *ip-address* **Example:** `Device(config-if)# vrrp 1 ip 10.0.1.20`	Enables VRRP on an interface and identifies the IP address of the virtual device.
Step 7	**end** **Example:** `Device(config-if)# end`	Returns to privileged EXEC mode.

Create and customize a VRRPv3 group

To create a VRRP group, perform this task. Steps 6 to 14 denote customizing options for the group, and they are optional:

Procedure

	Command or Action	Purpose
Step 1	**enable** **Example:** `Device> enable`	Enables privileged EXEC mode. Enter your password if prompted.
Step 2	**configure terminal** **Example:** `Device# configure terminal`	Enters global configuration mode.
Step 3	**fhrp version vrrp v3** **Example:** `Device(config)# fhrp version vrrp v3`	Enables the ability to configure VRRPv3 and VRRS.
Step 4	**interface** *type number* **Example:**	Enters interface configuration mode.

	Command or Action	Purpose
	`Device(config)# interface GigabitEthernet 0/0/0`	
Step 5	**vrrp** *group-id* **address-family** {**ipv4** | **ipv6**} **Example:** `Device(config-if)# vrrp 3 address-family ipv4`	Creates a VRRP group and enters VRRP configuration mode.
Step 6	**address** *ip-address* [**primary** | **secondary**] **Example:** `Device(config-if-vrrp)# address 100.0.1.10 primary`	Specifies a primary or secondary address for the VRRP group. **Note** VRRPv3 for IPv6 requires that a primary virtual link-local IPv6 address is configured to allow the group to operate. After the primary link-local IPv6 address is established on the group, you can add the secondary global addresses.
Step 7	**description** *group-description* **Example:** `Device(config-if-vrrp)# description group 3`	(Optional) Specifies a description for the VRRP group.
Step 8	**match-address** **Example:** `Device(config-if-vrrp)# match-address`	(Optional) Matches secondary address in the advertisement packet against the configured address. **Note** Secondary address matching is enabled by default.
Step 9	**preempt delay minimum** *seconds* **Example:** `Device(config-if-vrrp)# preempt delay minimum 30`	(Optional) Enables preemption of lower priority primary device with an optional delay. **Note** Preemption is enabled by default.
Step 10	**priority** *priority-level* **Example:** `Device(config-if-vrrp)# priority 3`	(Optional) Specifies the priority value of the VRRP group. The priority of a VRRP group is 100 by default.
Step 11	**timers advertise** *interval* **Example:** `Device(config-if-vrrp)# timers advertise 1000`	(Optional) Sets the advertisement timer in milliseconds. The advertisement timer is set to 1000 milliseconds by default.

	Command or Action	Purpose
Step 12	**vrrpv2** **Example:** `Device(config-if-vrrp)# vrrpv2`	(Optional) Enables support for VRRPv2 configured devices in compatibility mode.
Step 13	**vrrs leader** *vrrs-leader-name* **Example:** `Device(config-if-vrrp)# vrrs leader leader-1`	(Optional) Specifies a leader's name to be registered with VRRS and to be used by followers. **Note** A registered VRRS name is unavailable by default.
Step 14	**shutdown** **Example:** `Device(config-if-vrrp)# shutdown`	(Optional) Disables VRRP configuration for the VRRP group. **Note** VRRP configuration is enabled for a VRRP group by default.
Step 15	**end** **Example:** `Device(config)# end`	Returns to privileged EXEC mode.

Configure the delay period before FHRP client initialization

To configure the delay period before the initialization of all FHRP clients on an interface, perform this task:

Procedure

	Command or Action	Purpose
Step 1	**enable** **Example:** `Device> enable`	Enables privileged EXEC mode. Enter your password if prompted.
Step 2	**configure terminal** **Example:** `Device# configure terminal`	Enters global configuration mode.
Step 3	**fhrp version vrrp v3** **Example:** `Device(config)# fhrp version vrrp v3`	Enables the ability to configure VRRPv3 and VRRS.

	Command or Action	Purpose
Step 4	**interface** *type number* **Example:** `Device(config)# interface GigabitEthernet 0/0/0`	Enters interface configuration mode.
Step 5	**fhrp delay** {[**minimum** | **reload**] *seconds*} **Example:** `Device(config-if)# fhrp delay minimum 5`	Specifies the delay period for the initialization of FHRP clients after an interface comes up. The range is 0-3600 seconds.
Step 6	**end** **Example:** `Device(config)# end`	Returns to privileged EXEC mode.

Track an IPv6 object using VRRPv3

To track an IPv6 object using VRRPv3, perform this task.

Procedure

	Command or Action	Purpose
Step 1	**enable** **Example:** `Device> enable`	Enables privileged EXEC mode. Enter your password if prompted.
Step 2	**configure terminal** **Example:** `Device# configure terminal`	Enters global configuration mode.
Step 3	**fhrp version vrrp v3** **Example:** `Device(config)# fhrp version vrrp v3`	Enables you to configure VRRPv3 and Virtual Router Redundancy Service (VRRS) on a device. **Note** When VRRPv3 is in use, VRRPv2 is unavailable.
Step 4	**interface** *type number* **Example:** `Device(config)# interface GigabitEthernet 0/0/0`	Specifies an interface and enters interface configuration mode.

	Command or Action	Purpose
Step 5	**vrrp** *group-id* **address-family ipv6** **Example:** `Device(config-if)# vrrp 1 address-family ipv6`	Creates a VRRP group for IPv6 and enters VRRP configuration mode.
Step 6	**track** *object-number* **decrement** *number* **Example:** `Device(config-if-vrrp)# track 1 decrement 20`	Configures the tracking process to track the state of the IPv6 object used by the VRRPv3 group. VRRP on the interface then registers with the tracking process to be informed of any changes to the IPv6 object on the VRRPv3 group. If the tracked IPv6 object state configured on the interface goes down, the priority of the VRRP group is reduced by 20.
Step 7	**end** **Example:** `Device(config-if-vrrp)# end`	Returns to privileged EXEC mode.

Monitor VRRP

The commands in this section can be used to monitor VRRP.

Example: Monitor VRRP status, configuration, and statistics details

This is a sample output of the status, configuration, and statistics details for a VRRP group:

```
Device# show vrrp detail

 GigabitEthernet1/0/1 - Group 3 - Address-Family IPv4
  Description is "group 3"
  State is MASTER
  State duration 53.901 secs
  Virtual IP address is 100.0.1.10
  Virtual MAC address is 0000.5E00.0103
  Advertisement interval is 1000 msec
  Preemption enabled, delay min 30 secs (0 msec remaining)
  Priority is 100
  Master Router is 10.21.0.1 (local), priority is 100
  Master Advertisement interval is 1000 msec (expires in 832 msec)
  Master Down interval is unknown
  VRRPv3 Advertisements: sent 61 (errors 0) - rcvd 0
  VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
  Group Discarded Packets: 0
    VRRPv2 incompatibility: 0
    IP Address Owner conflicts: 0
    Invalid address count: 0
    IP address configuration mismatch : 0
    Invalid Advert Interval: 0
    Adverts received in Init state: 0
```

```
    Invalid group other reason: 0
  Group State transition:
    Init to master: 0
    Init to backup: 1 (Last change Sun Mar 13 19:52:56.874)
    Backup to master: 1 (Last change Sun Mar 13 19:53:00.484)
    Master to backup: 0
    Master to init: 0
    Backup to init: 0
```

Example: Monitor VRRP IPv6 object tracking

These examples show how to verify the VRRP IPv6 object tracking process configuration:

```
Device# show vrrp

GigabitEthernet0/0/0 - Group 1 - Address-Family IPv4
  State is BACKUP
  State duration 1 mins 41.856 secs
  Virtual IP address is 172.24.1.253
  Virtual MAC address is 0000.5E00.0101
  Advertisement interval is 1000 msec
  Preemption enabled
  Priority is 80 (configured 100)
    Track object 1 state Down decrement 20
  Master Router is 172.24.1.2, priority is 100
  Master Advertisement interval is 1000 msec (learned)
  Master Down interval is 3609 msec (expires in 3297 msec)

Device# show track ipv6 route brief

Track Type        Instance                   Parameter        State Last Change
601   ipv6 route  3172::1/32                 metric threshold Down  00:08:55
602   ipv6 route  3192:ABCD::1/64            metric threshold Down  00:08:55
603   ipv6 route  3108:ABCD::CDEF:1/96       metric threshold Down  00:08:55
604   ipv6 route  3162::EF01/16              metric threshold Down  00:08:55
605   ipv6 route  3289::2/64                 metric threshold Down  00:08:55
606   ipv6 route  3888::1200/64              metric threshold Down  00:08:55
607   ipv6 route  7001::AAAA/64              metric threshold Down  00:08:55
608   ipv6 route  9999::BBBB/64              metric threshold Down  00:08:55
611   ipv6 route  1111::1111/64              reachability     Down  00:08:55
612   ipv6 route  2222:3333::4444/64         reachability     Down  00:08:55
613   ipv6 route  5555::5555/64              reachability     Down  00:08:55
614   ipv6 route  3192::1/128                reachability     Down  00:08:55
```